

1. Relations & Functions

A **relation** is a connection between elements of two sets. A **function** is a special type of relation where each input has **exactly one output**.

1. Types of Relations

Let A and B be two sets. A **relation R from A to B** is a subset of $A \times B$ (the Cartesian product).

Common types of relations:

1. **Reflexive Relation:**

A relation R on set A is reflexive if $(a, a) \in R$ for every $a \in A$.

Example: Equality ($=$) is reflexive because $a = a$.

2. **Symmetric Relation:**

R is symmetric if $(a, b) \in R \implies (b, a) \in R$.

Example: "Is a sibling of" is symmetric.

3. **Transitive Relation:**

R is transitive if $(a, b) \in R$ and $(b, c) \in R \implies (a, c) \in R$.

Example: "Is ancestor of" is transitive.

4. **Equivalence Relation:**

A relation that is **reflexive, symmetric, and transitive**.

Example: Congruence modulo n .

5. **Partial Order:**

A relation that is **reflexive, antisymmetric, and transitive**.

Example: $\leq$ on real numbers.

2. Functions (Domain, Range, Co-domain)

A function $f : A \rightarrow B$ is a relation where **every element in set A has exactly one image in set B** .

- **Domain:** Set of all possible inputs (elements of A).
- **Co-domain:** Set of all possible outputs (set B) **defined** by the function.
- **Range (or Image):** Set of actual outputs $f(a)$ for all $a \in A$.

Example: $f(x) = x^2, f : \mathbb{R} \rightarrow \mathbb{R}$

- Domain = $\mathbb{R}$
 - Co-domain = $\mathbb{R}$
 - Range = $[0, \infty)$
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3. Types of Functions

1. **One-to-One (Injective):**

$f(a_1) = f(a_2) \implies a_1 = a_2$.

No two elements of domain map to the same element of range.

2. Onto (Surjective):

Every element of co-domain has a pre-image in domain.

Range = Co-domain.

3. Bijective:

Function is both injective and surjective.

Every element of the co-domain has **exactly one** pre-image.

4. Constant Function:

Maps all elements of the domain to a single element in co-domain.

5. Identity Function:

$f(x) = x$ for all x in domain.

6. Polynomial, Exponential, Logarithmic, Trigonometric Functions (exam may ask examples).

4. Composition of Functions

If $f : A \rightarrow B$ and $g : B \rightarrow C$, the **composition** $g \circ f$ is defined as:

$$(g \circ f)(x) = g(f(x)) \quad \text{for all } x \in A$$

Properties:

- Not always commutative: $g \circ f \neq f \circ g$ in general.
 - Associative: $h \circ (g \circ f) = (h \circ g) \circ f$
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5. Invertible Functions

A function $f : A \rightarrow B$ is **invertible** if there exists a function $f^{-1} : B \rightarrow A$ such that:

$$f^{-1}(f(x)) = x \quad \text{and} \quad f(f^{-1}(y)) = y$$

- Only **bijective functions** are invertible.
 - Graphically, f and f^{-1} are reflections across $y = x$.
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✓ Quick Tips for Exams:

- Memorize **domain**, **co-domain**, **range** examples.
- Always check if function is **one-to-one** and **onto** before finding inverse.
- Use **diagrams** for relations: helpful for reflexive, symmetric, transitive.
- Practice **composition** problems step by step.